

# Md Rahatul Ashakin

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## Research Focus

My research focuses on quantum machine learning and advanced AI for solving challenges in healthcare, biology, and secure data systems. I am passionate about developing innovative, interdisciplinary methods that enhance prediction, diagnosis, and data-driven discovery while ensuring strong security. I seek to collaborate across domains and apply new technologies to impactful problems in science and society.

## Research and Professional Experience

**Data Science Intern**, Tekurai Inc., San Antonio, Texas (remote, Queens - NY) *May 2024 to Present*

- Built transformer and CNN pipelines for clinical text and audio, analyzed clinician and patient interactions for decision support.
- Processed more than five hundred thousand healthcare records with Python and produced interactive Power BI dashboards for operations and care insights.
- Prototyped privacy aware blockchain workflows for secure data exchange in drug discovery across multiple institutions.
- Quantum machine learning with Qiskit, including hybrid variational models and quantum guided hyperparameter search that improved internal accuracy and AUC over classical baselines.

**Program Manager**, ShopUp, Dhaka, Bangladesh *Feb 2022 to Sep 2022*

- Led analytics for B2C and B2B operations with forecasting and KPI reporting; delivered Tableau and Power BI dashboards used to reduce returns and improve throughput.
- Partnered with operations, product, and engineering to define a KPI taxonomy and the supporting SQL/ETL pipelines; automated weekly executive dashboards and cohort analyses to speed up decision cycles.

**Project Manager**, Icon Information Systems Ltd., Dhaka, Bangladesh *Jun 2020 to Feb 2022*

- Directed cross-team reporting and SQL data workflows for enterprise clients; coordinated agile ceremonies and risk tracking.
- Produced project plans and stakeholder updates, organized sprints and stand-ups, and prepared UAT and release checklists to ensure on-time, auditable deliveries.

**Assistant Project Manager**, Millennium Information Solution Ltd., Dhaka, Bangladesh *Nov 2018 to May 2020*

- Supported delivery of HRIS and financial systems with user acceptance testing, reliability checks, and transaction safety reviews for regulated stakeholders.

## Education

**M.S., Information Technology**, Washington University of Science and Technology (WUST), Alexandria, Virginia *Oct 2022 to Mar 2024*

**B.S., Computer Science and Engineering**, North South University, Dhaka, Bangladesh *Sep 2013 to Aug 2018*

## Publications & Scholarly Contributions

### Book Chapter (forthcoming)

- **Ashakin, M. R.** (2026). Chapter 5: *Current Tools and Techniques of Artificial Intelligence in Healthcare*. In D. Roosan (Ed.), *Quantum Computing in Medicine: Transforming Drug Discovery, Protein Folding, and Genomics Research through Artificial Intelligence Integration*. Elsevier. ISBN: 978-0-443-34193-9. *CRedit*: Conceptualization; Methodology; Writing: original draft; Visualization; Writing: review & editing.

### Under Review (Journal)

- **Ashakin, M. R.**, Stuetzle, C., Khan, R., Nirzhor, S., Roosan, D. *Quantum Annealing of Financial Sentiment Analysis by Optimizing Large Language Model Hyperparameters*. Under review at *Financial Innovation (SpringerOpen)*. *CRedit*: Lead: Conceptualization; Methodology; Software; Formal analysis; Visualization; Writing: original draft. Supporting: Data curation; Writing: review & editing. *Resources*: Code: [GitHub repository](#).

### Refereed Conference Papers

- Roosan, D., **Ashakin, M. R.**, Khan, R., Hasiba Mashed Khan, Khou, T., Carnasciali, M. I., Haider, M. R. (2025). *Variational Quantum Circuits for Molecular Classification Using Graph Neural Network*. In: IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC 2025). *CRedit*: Methodology; Software; Formal analysis; Visualization; Writing: review & editing.
- Roosan, D., Khan, R., **Ashakin, M. R.**, Khou, T., Nirzhor, S., Haider, M. R. (2025). *Quantum Variational Transformer Model for Enhanced Cancer Classification*. IEAI 2025, IOS Press, *Advances in Transdisciplinary Engineering*. *CRedit*: Methodology; Software; Validation; Formal analysis; Writing: review & editing.
- Roosan, D., Khan, R., **Ashakin, M. R.**, Khou, T. (2025). *Adaptive Multimodal Artificial Intelligence with Liquid Neural Network for Edge Computing-Based Augmented Reality*. IEAI 2025, IOS Press, *Advances in Transdisciplinary Engineering*. *CRedit*: Conceptualization; Methodology; Software; Visualization; Writing: review & editing.
- Roosan, D., Khou, T., Pham, B., Li, Y., **Ashakin, M. R.**, Hasiba Mashed Khan, Khan, R., Haider, M. R. (to appear, Nov 2025). *Quantum AI Enhanced Blockchain Security for Drug Discovery*. *Proceedings of ICAI 2025, Springer LNEE (to appear)*. *CRedit*: Conceptualization; Methodology; Writing: review & editing.

### Journal Articles

- **Ashakin, M. R.**, Bhuyian, M. S., Hosain, M. R., Deya, R. S., Hasan, S. E. (2024). *Transforming to Smart Healthcare: AI Innovations for Improving Affordability, Efficiency, and Accessibility*. *Pathfinder of Research*, 2024. *CRedit*: Lead: conceptualization; methodology; writing: original draft; visualization; writing: review & editing.
- Hossain Ifty, **Ashakin, M. R.**, S. M., Hossain, B. Islam Tusher, M. B., Haque, R. (2023). *Adoption of IoT in Agriculture — Systematic Review*. *Applied Agriculture Sciences*, 2023. *CRedit*: Writing: review & editing; literature curation; visualization.
- **Ashakin, M. R.**, Hossain, B., Hossain Ifty, S. M. (2024). *Enhancing Digital Signal Processing: Obstacles and Advancements in FPGA and ASIC Hardware Implementations*. *Pathfinder of Research*, 2024. *CRedit*: Lead: conceptualization; writing: original draft; methodology; visualization; writing: review & editing.
- Hossain Ifty, S. M., **Ashakin, M. R.**, Hossain, B., Afrin, S., Sattar, A., Chowdhury, R., Tusher, M. I., Bhowmik, P. K., Mia, M. T., Islam, T., Tufael, M., Sunny, A. R. (2023). *IOT-Based Smart Agriculture in Bangladesh: An Overview*. *Applied Agriculture Sciences*, 2023. *CRedit*: Writing: review & editing; visualization; literature curation.
- Salam, M. T., Bari, K. F., Rahman, M. M., Gafur, D. M. M., Faruk, M. O., Akter, K., Irin, F., **Ashakin, M. R.**, Shaikat, T. A., Das, A. C., Tufael, M., Mithun, M. M., Sunny, A. R. (2024). *Emergence of Antibiotic-Resistant Infections in ICU Patients*. *Journal of Angiotherapy*, 2024. *CRedit*: Writing: review & editing; visualization.

- **Ashakin, M. R.**, Hasan, M. R. K., Urbi, S. R. C. (2024). *Impact of Internet of Things (IoT) on Healthcare in Transforming Patient Care and Overcoming Operational Challenges*. *Journal of Angiotherapy*, 2024. *CRedit*: Lead: conceptualization; writing: original draft; visualization; writing: review & editing.
- **Ashakin, M. R.**, Hasan, M. R. K., Urbi, S. R. C. (2024). *AI-Powered Methods for Smarter Decisions in Automated Machine Learning in Business Analytics*. *Pathfinder of Research*, 2024. *CRedit*: Lead: conceptualization; writing: original draft; methodology; visualization; writing: review & editing.
- Hossain Ifty, S. M., Irin, F., Shovon, M. S. S., Amjad, M. H. H., Bhowmik, P. K., Ahmed, R., **Ashakin, M. R.**, Hossain, B., Mushfiq, M., Sattar, A., Chowdhury, R., Sunny, A. R. (2024). *Advancements, Applications, and Future Directions of Artificial Intelligence in Healthcare*. *Journal of Angiotherapy*, 2024. *CRedit*: Writing: review & editing; visualization.

## Skills

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|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methods &amp; Algorithms</b>      | Transformers · CNNs · GNNs · multimodal learning · optimization for robustness · model calibration · privacy-aware modeling                  |
| <b>Data &amp; Experimentation</b>    | Dataset curation · experimental design · cross-validation · statistical testing · ablation studies · error analysis · reproducible pipelines |
| <b>Software &amp; Systems</b>        | Python · SQL · Git · Linux · Jupyter · Pandas · NumPy · scikit-learn · PyTorch · Hugging Face                                                |
| <b>Visualization &amp; Reporting</b> | Matplotlib · Tableau · Power BI · results dashboards · academic writing with LaTeX/Overleaf                                                  |
| <b>Computing Platforms</b>           | GPUs · cloud · edge/IoT deployment                                                                                                           |

## Selected Research Projects (open-source)

- **IBD-Quantum-ESM**. Quantum-informed *IBD* modeling pipeline (optimization with classical + quantum methods). *Resources*: Code: [GitHub](#) · [README](#) · [Notebooks](#).
- **Quantum Optimization Sandbox**. Lightweight scaffold to *repurpose* ML experiments with QAOA/VQE and classical baselines. *Resources*: Code: [GitHub](#) · [README](#) · [Notebooks](#).

## Online Courses & Certifications (selected)

- [Microsoft Power BI Data Analyst Specialization](#) · **Microsoft**
- [Google Advanced Data Analytics](#) · **Google**
- [Google Business Intelligence](#) · **Google**
- [Google Data Analytics](#) · **Google**
- [SAS Statistical Business Analyst](#) · **SAS**
- [Project Management & Other Tools for Career Development](#) · **UC Irvine**

## Academic Service

**Research Programmer**, Merrimack College, North Andover, Massachusetts (Remote, Queens - NY) *Sep 2024 to Dec 2024*

- Supported *DSE 4900: Data Science Capstone (Fall 2024)* and *DSE 3010: Machine Learning for Data Science (Fall 2024)*; coordinated labs and weekly research clinics, and supervised capstone theses end-to-end (scoping, dataset curation, modeling plans, final reports/presentations).
- Instituted reproducible research practice (clean repos, environment pinning, experiment tracking) with rigorous evaluation (ablations, error analysis, checks for leakage/imbalance/overfitting), enabling teams to achieve *up to ~50% relative accuracy gains* over course baselines where applicable.